

Principles of Software Design

Hospital Management System (HMS)

Project's Team

No	Name	ID
[1]	Ahmed Mostafa	2023/03479
[2]	Abdelrahman Hesham	2023/06186
[3]	Raafat Fayez	2023/07237
[4]	Michael Hisham	2023/01857
[5]	Moataz Mohamed	2023/10288

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Description

A Hospital Management System (HMS) is a Web Application to help hospitals manage their daily operations in an organized and efficient way. The system provides a centralized database where all important information related to the hospital is stored and managed. This includes patient details, appointment scheduling, medical records, billing information, medication stock, and staff data.

The HMS helps reduce paperwork and manual work by digitizing hospital processes. Doctors can easily access patient medical histories and update treatment records, while administrators can manage hospital resources, staff information, and system operations. Patients can use the system to schedule appointments, view their medical information, and communicate with doctors.

The system supports multiple types of users, such as Doctors, Administrators, and Patients, and each user has specific roles and permissions within the system. Doctors check medical records and previous treatments, administrators handle system management and hospital operations, and patients can access their personal health information and appointment details.

Overall, the Hospital Management System (HMS) improves efficiency, reduces errors, saves time, and enhances the quality of healthcare services by ensuring that all hospital information is organized, secure, and easily accessible.

Modules/services

Admin Panel Modules/services

1. Dashboard Module

The Dashboard module provides the administrator with a quick overview of the hospital system. It displays important statistics such as the total number of patients, doctors, appointments. It may also show recent activities, upcoming appointments, and system notifications. This helps the admin monitor the overall status of the hospital operations from one place.

2. Patient Management Module

This module allows the administrator to manage all patient records in the system. The admin can add new patients, update patient information, view patient profiles, and delete records if needed. It ensures that all patient data is stored correctly and can be accessed easily when required.

3. Doctor Management Module

The Doctor Management module allows the admin to manage doctor accounts and information. The admin can add new doctors, update their details, assign departments or specializations, and manage their availability schedules. This helps organize the hospital's medical staff efficiently.

4. Staff Management Module

The Staff Management module allows the admin to manage hospital employees such as nurses, receptionists, and other staff members. The admin can add staff records, update their information, assign roles, and manage their work schedules.

5. Medication / Pharmacy Management Module

This module allows the admin to manage medicines and pharmacy inventory. The admin can add new medications, update stock quantities, check medicines stock, and check expired medicines. It helps maintain accurate records of all medical supplies.

6. Billing and Payment Management Module

The Billing module allows the admin to manage financial records related to hospital services. The admin can view patient bills, generate invoices, confirm payments, and track financial transactions. This ensures transparency and accurate financial management.

7. User and Role Management Module

The User Management module allows the admin to control system users and their permissions. The admin can create accounts for doctors and staff, assign roles, and control what each user can access within the system. This helps maintain system security and proper access control.

Patient Services/Modules

1. Patient Registration Module

The Patient Registration module allows new patients to create an account in the system. Patients can enter their personal information such as name, age, contact details, and basic medical information also the ability to submit recent Medical examinations. This module ensures that each patient has a unique profile stored in the system database.

2. Profile Management Module

This module allows patients to view and update their personal information. Patients can edit details such as phone number, address, or other personal data. It helps keep patient information accurate and up to date.

3. Appointment Booking Module

The Appointment Booking module allows patients to schedule appointments with doctors. Patients can view available doctors, select a suitable date and time, and confirm their appointment. This module helps patients easily arrange visits without needing to go to the hospital in person.

4. Medical Records Viewing Module

The Medical Records module allows patients to access their medical history. Patients can view diagnoses, treatment details, prescriptions, and test results recorded by doctors. This helps patients stay informed about their health condition.

5. Prescription Viewing Module

This module allows patients to view prescriptions given by doctors after consultations. Patients can check the medications prescribed, dosage instructions, and other medical recommendations.

6. Billing and Payment Module

The Billing module allows patients to view their hospital bills and payment details. Patients can check charges for consultations, treatments, and medications. In some systems, patients can also pay their bills online through this module.

7. Notifications Module

The Notifications module sends reminders and updates to patients. Patients can receive notifications about upcoming appointments, prescription updates, or important hospital announcements.

Doctor Modules/services

1. Doctor Dashboard Module

The Doctor Dashboard provides doctors with a quick overview of their daily activities. It shows important information such as upcoming appointments, recent patient visits, notifications, and system updates. This helps doctors quickly see their Appointment.

2. Patient Records Management Module

This module allows doctors to view patient profiles and medical histories. Doctors can access important patient information such as previous diagnoses, treatments, test

results, and prescriptions. This helps doctors understand the patient's health condition before providing treatment.

3. Appointment Management Module

The Appointment Management module allows doctors to view their scheduled appointments with patients. Doctors can see appointment details such as patient name, date, and time. They may also update the appointment status, such as marking it as completed or canceled.

4. Diagnosis and Treatment Module

This module allows doctors to record diagnoses and treatment details for patients. During or after a consultation, doctors can enter notes about the patient's condition, recommended treatments, and medical advice. These records are stored in the patient's medical history.

5. Prescription Management Module

The Prescription module allows doctors to create and manage prescriptions for patients. Doctors can enter the name of medications, dosage instructions, and treatment duration. The prescriptions are then stored in the system and can be viewed by the patient.

6. Medical Reports and Test Results Module

This module allows doctors to view and update patient medical reports and laboratory test results. Doctors can analyze the results and add notes or recommendations based on the findings.

General Modules/Services

1. Authentication and Login Module

The Authentication module manages user login and security. It allows users to log into the system using their username and password. The system verifies the user's identity and provides access based on their role (Admin, Doctor, or Patient). This module ensures that only authorized users can access the system.

2. User Profile Management Module

This module allows users to view and manage their personal profile information. Users can update details such as contact information, password, and other personal data. It helps keep user information accurate and secure.

3. Notification and Alert Module

The Notification module sends important alerts and reminders to users. These notifications may include appointment reminders, system updates, prescription notifications, or important hospital announcements. This helps keep users informed about important activities.

4. Search Module

The Search module allows users to quickly find information in the system. Users can search for patients, doctors, appointments, medical records, or medications depending on their access level. This improves efficiency and saves time when accessing data.

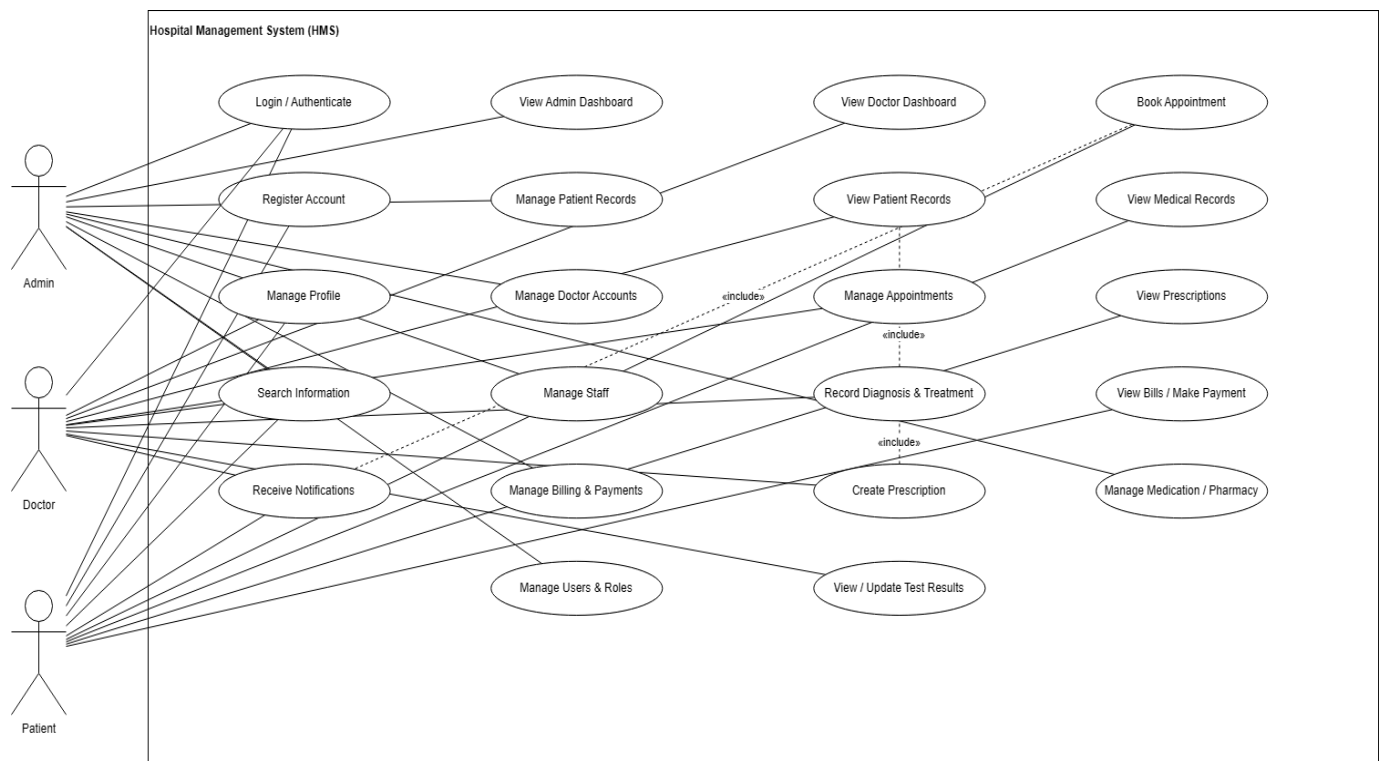
5. Security and Access Control Module

The Security module manages system protection and user permissions. It controls what each user can access in the system based on their role. This helps protect sensitive medical information and prevents unauthorized access.

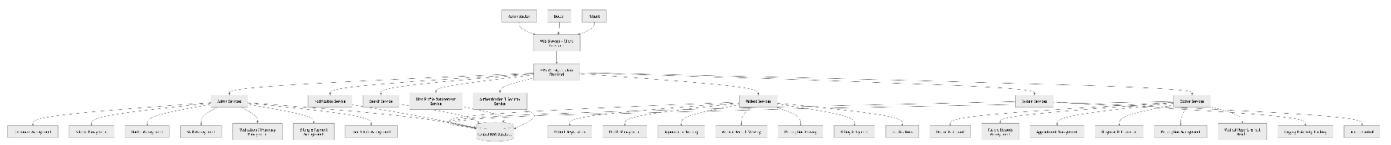
6. Logging and Activity Tracking Module

This module records system activities and user actions such as logins, updates, and data changes. It helps administrators monitor system usage and detect any unusual activity.

Use Case Diagram



System Architecture Diagram



ERD

